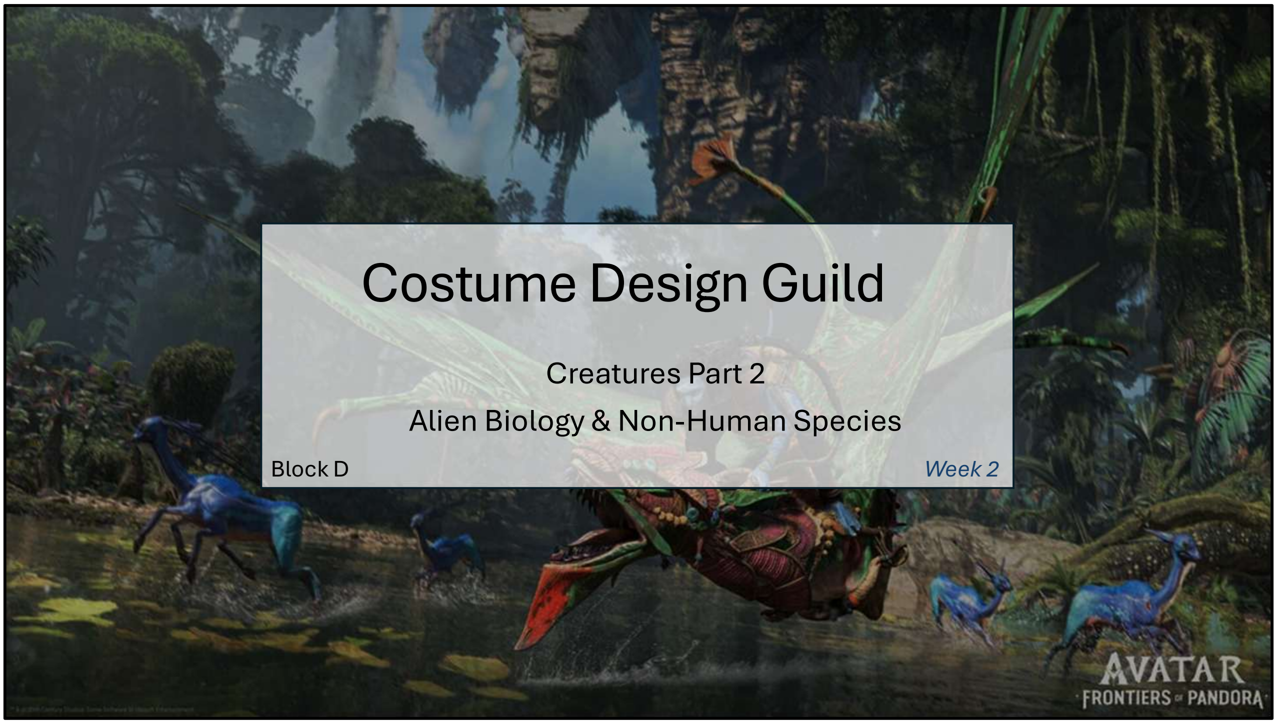

Costume Design Guild
Creatures Part 2
Alien Biology & Non-Human Species
Block D
Week 2
AVATAR
FRONTIERS OF PANDORA

Welcome back!

New format this block:

- 20 min: Theory/Discussion
- 20 min: Activity
- 20 min: Student Work Review (voluntary)

Block D runs a bit differently than Block C.
Twenty minutes theory, twenty minutes activity, and twenty minutes voluntary work review.
I don’t think the timing will be exactly 20 minutes but we’ll see during this session.
These sessions are no longer for my data gathering as I’ve finished that but I’ve kept the questions as you guys liked them (and I do too)

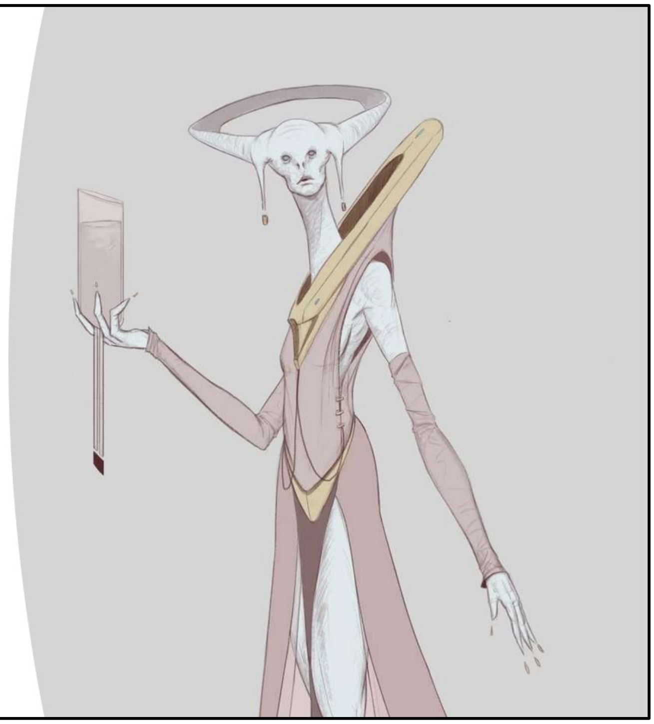
Agenda

Foundations (quick)

Theory
- Biology → Material Needs
- Environmental Adaptations
- Game Examples

Activity
- Analyse 2 alien species
- Identify costume/material needs based on biology

Work Review
- Voluntary sharing
- Group feedback and discussion

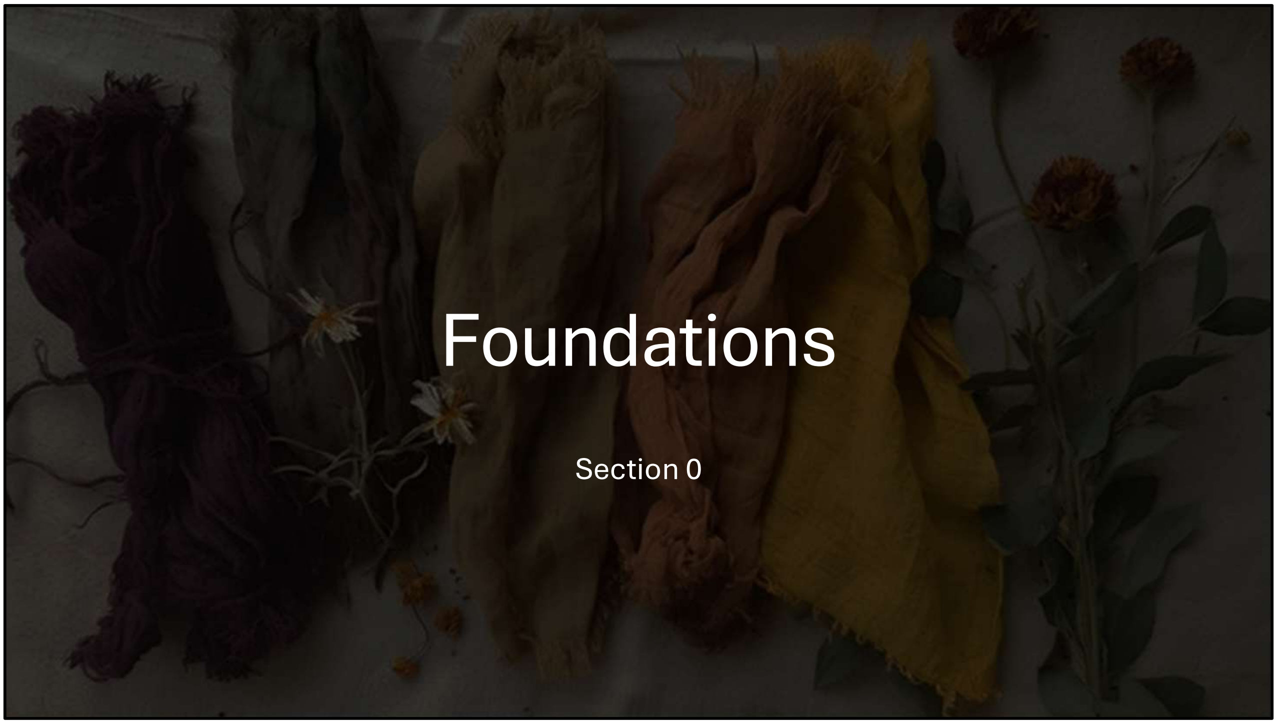


Quick foundations refresher first — reminder if you were in Block C, intro if you're new.
Then alien costume theory, a species analysis activity, and work review at the end.

Check-in question

- Have you designed any non-human species before?
- What was the biggest challenge with their costume?

Before we start — have you designed clothing for non-human species before? So creatures?
What was the hardest part with their costume?



Lets get started with the foundations

Core Concepts Reminder

- If you joined Block C guilds, this is a quick refresher.
- If you're new, these are the foundational principles we'll build on today.

Core concepts:

- Visual language
- Layers
- Material behaviour
- Historical grounding

These four principles ran through all of Block C.
Today we're applying them to aliens.
So this will be a quick warm-up if you know them, quick intro if you don't.

Visual language

What influences visual language:

- **Silhouette** = Shape/outline
  (most recognizable from distance)
- **Materials** = How it moves/behaves
  (weight, drape, texture)
- **Construction** = How it's made
  (seams, structure, fit)
- **Details** = Embellishment, closures, trim

Why this matters: Same character archetype, different silhouettes = completely different read

The four things that make a costume readable are: silhouette, materials, construction, details.
Silhouette is what registers first, even from a distance.
Materials and their properties is what influence the way the garment moves
Construction is how its made
And the details are what makes a garment interesting

- This matters for game design because the same character archetype reads completely when wearing different silhouettes.

For example, Ellie from the last of us on the right wearing her normal outfit. And below a space suit. If you know her story it kinda matches but it breaks immersion in a survival world. Its not practical.

Layers

- **Underlayers** → Foundation, structure (chemise, bodysuit, corset)
- **Mid layers** → Main garment (tunic, dress, armour)
- **Outer layers** → Weather, status, decoration (cloak, jacket)

Why this matters: Affects how cloth behaves in simulation

rigid underlayer + draped outer = two-layer approach

Characters move differently with multiple layers

Most clothing is built in layers so under, mid, and outer layers

--For cloth simulation this matters because a rigid underlayer changes how the outer layer drapes
For example a women wearing hoop skirts and a corset will move differently than a women in athletic stretchy clothing.
They will have completely different ranges of movement and the layers of clothing will move as a separate shape on the body or WITH the body.

Material behaviour

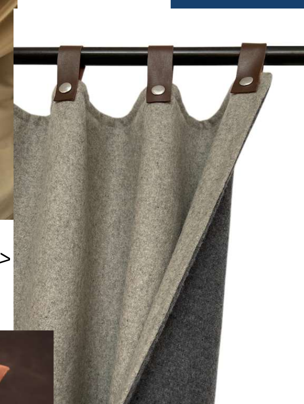
Silk ≠ Wool ≠ Leather

Different materials = completely different:

- **Drape** (how it falls)
- **Weight** (how gravity affects it)
- **Stretch** (does it return to shape?)
- **Texture** (smooth, rough, fuzzy)
- **Movement** (stiff, fluid, bouncy)

For game artists: Different cloth simulation settings Different shader properties Different animation behaviour

< Silk



Wool >

< Leather

Silk, wool, leather all have a completely different drape due to, weight, stretch, texture and movement.
You could feel this if you attended the hands-on workshop.

--In game work it means different cloth sim settings, different shaders, different animation behaviour. And material is not just for decoration.

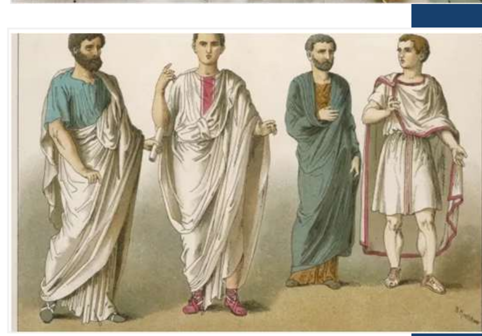
Historical grounding

Players/Viewers Have Expectations

Even in fantasy or sci-fi, historical grounding makes designs believable.

Why?

- Players **recognize** silhouettes and materials from real-world reference (even subconsciously)
- **Cultural coding** through media creates expectations *(elves = medieval, vampires = Victorian)*
- **Familiar** elements anchor the unfamiliar *(design feels more real when it references known logic)*



Players come with expectations even for sci-fi.
Elves feel medieval, vampires feel Victorian and that's cultural coding from media.
When you anchor alien design in something familiar, the unfamiliar parts land better.
That's why Block C was about historical periods. Not for accuracy, but for the visual vocabulary.
You can watch back the recordings of look at the slides if you want, they’re all in this teams channel.
Also, download everything before I leave buas and you cant access them anymore.

Now Let's Apply These Foundations

Today we're asking:

- What SILHOUETTES would alien biology create?
- What MATERIALS would they invent based on their needs?
- How would CONSTRUCTION adapt to non-human bodies?
- What DETAILS communicate their culture?

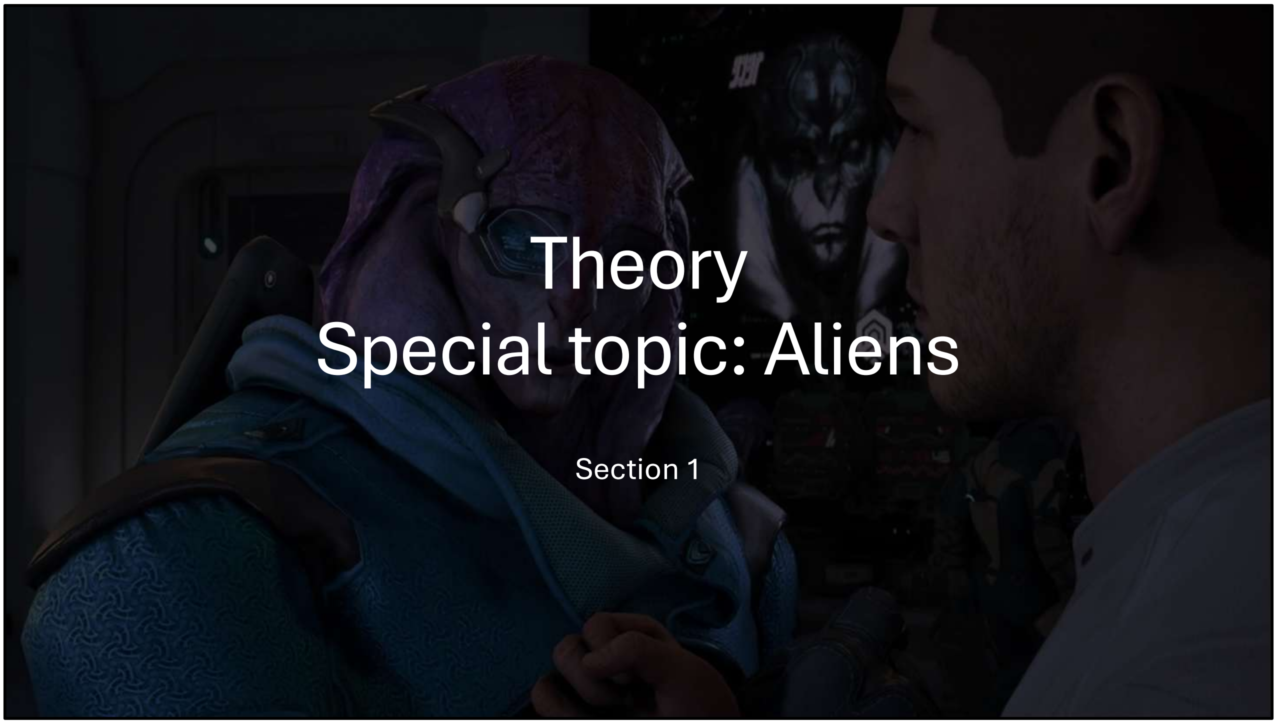
Today we'll be asking: what would this species actually invent for themselves? What silhouettes come from their biology, what materials from their environment, what construction from their body and what details from their culture?

Quick question

Have you designed for aliens?

Walk me through your thought process

So before we dive into that a quick question again
Have any of you designed for aliens specifically, not just fantasy creatures? Walk me through how you thought about it



With those foundations in mind, let's dive into alien species design.

Topic introduction

- What would a species with different biology actually INVENT for clothing?

Most game aliens wear '*human clothing with minor tweaks*'

But what if we thought deeper about:

- Their actual biological needs?
- Their environment?
- What materials they'd have access to?
- What problems clothing would solve for THEM?

Most game aliens wear human clothing with a texture swap and maybe a tail cutout.
The body changed but the costume logic didn't.
So if you actually ask what this species needs from clothing, what they'd have access to, what problem it solves for them — you get to something much more interesting and that's what we're talking about today

Key concepts

Different biology = Different material requirements

Human Example (Baseline):

- Warm-blooded → Need insulation in cold, cooling in heat
- Skin sweats → Need breathable fabrics
- No natural armour → Need protective materials
- Omnivorous → Can use animal AND plant materials

Lets start with humans as the baseline.
Warm-blooded, sweating skin, no natural armour, omnivorous.

Those four facts shaped everything we historically made for clothing.
Change anyone of them and the design requirements changes.

Key concepts

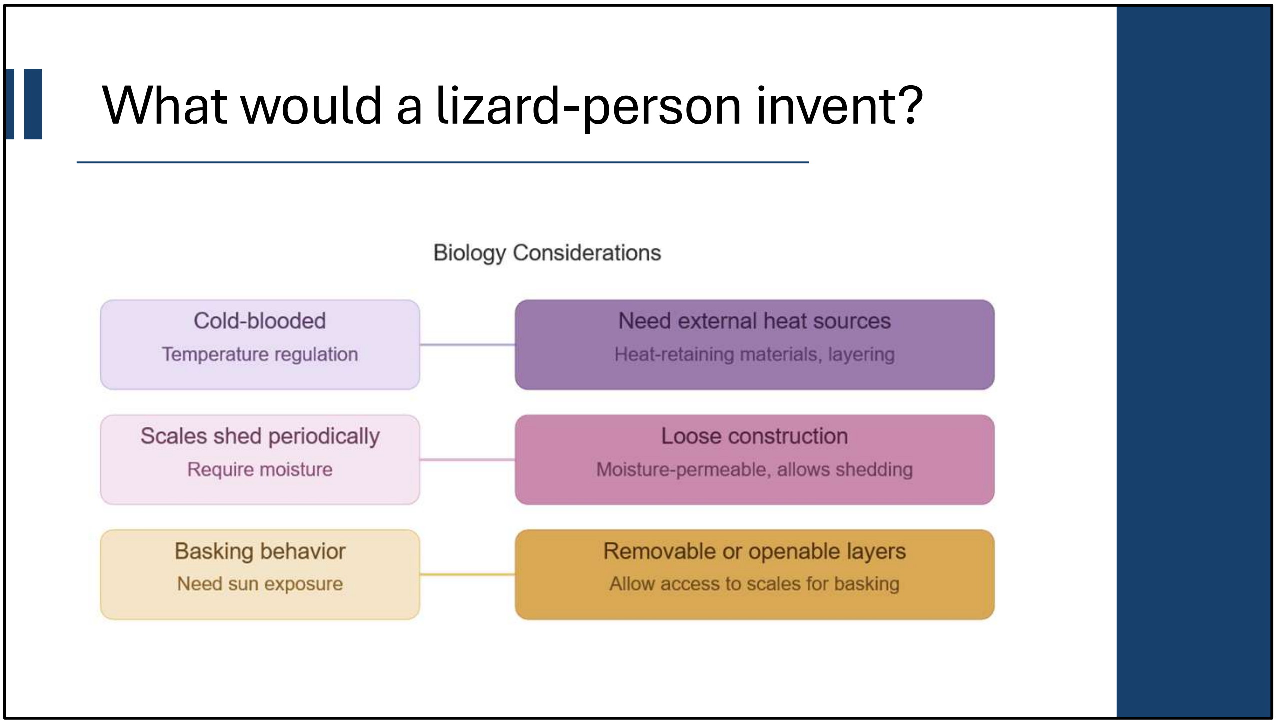
Different biology = Different material requirements

Non-Human Questions:

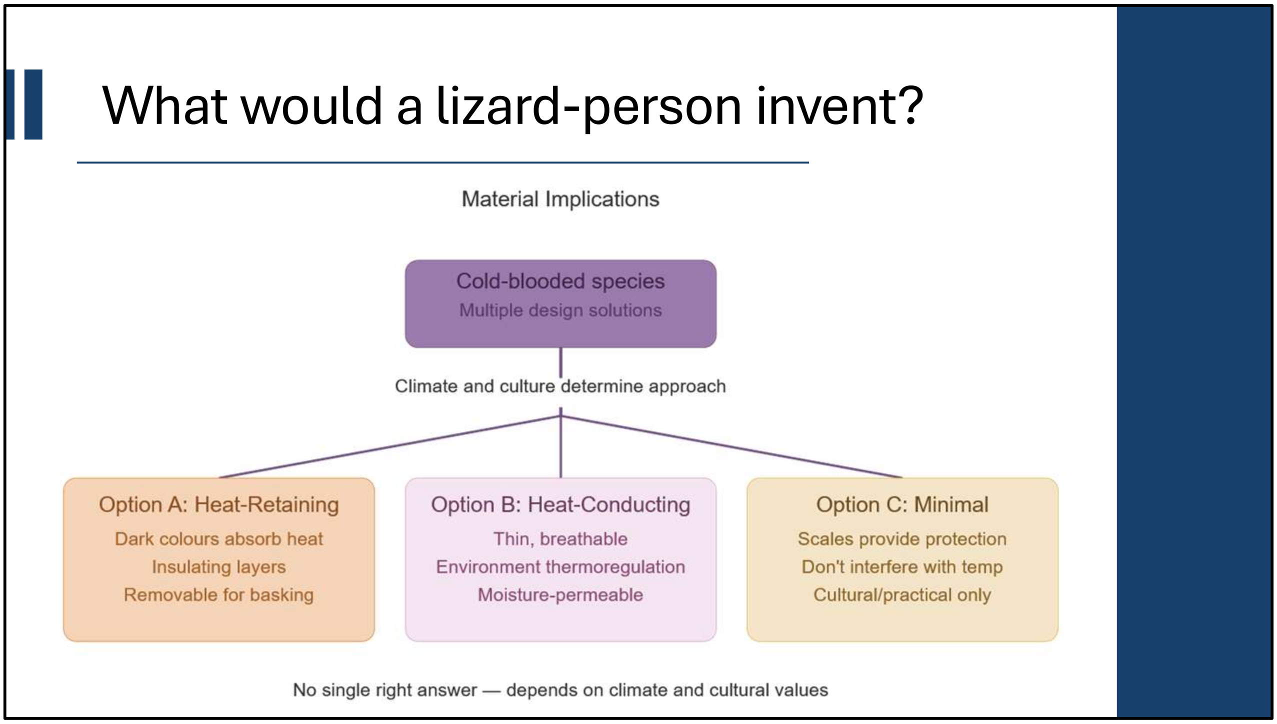
- Cold-blooded → Do they need heat-retaining OR heat-absorbing materials?
- Scales/fur/feathers → Do they even need fabric, or does it interfere?
- Natural armor → Is clothing for protection redundant?
- Aquatic → Does fabric impede their primary locomotion?

So for a non-human:
cold-blooded means different thermoregulation — do they need heat-retaining or heat-absorbtion?
Or Scales or fur might mean fabric interferes or not need anything at all
Natural armour might make protective clothing pointless.
And Aquatic movement might mean fabric creates drag.

But these aren't problems. You can see them as design prompts.



Here you have three biological facts, and their three design directions.
Cold-blooded might need external heat points or heat-retaining materials and layered clothing
Scales that shed might require moisture and would need a looser construction to allow this shedding
Or Basking behaviour would need sun exposure which would require removable or openable layers to allow for that



From those directions you get three material options.
Option A Heat-retaining: can be designed as dark colours, insulation, and removable pieces.
Option B Heat-conducting: can be designed as thin, thermoregulating, moisture-permeable materials or construction
And Option C Minimal: might have scales that already handle protection and temperature, so clothing is purely cultural.

So it really depends on the world and the culture.

What would a lizard-person invent?

There's no single "right" answer

It depends on THEIR world's climate and culture

And there's no single lizard-person.
Look at the range of real reptiles — a chameleon, snake and a gecko are all reptiles and completely different.
The answer always depends on the specific biology, the world's climate, and what the culture values.

What's available dictates what's possible

Desert Planet:

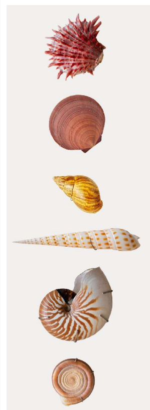
- Limited plant fibers
  → Possibly leather from local fauna, woven grasses if available
- Extreme temperatures
  → Need insulation AND cooling (layering system?)
- Water scarcity
  → Materials that don't need washing, protective from sun/sand

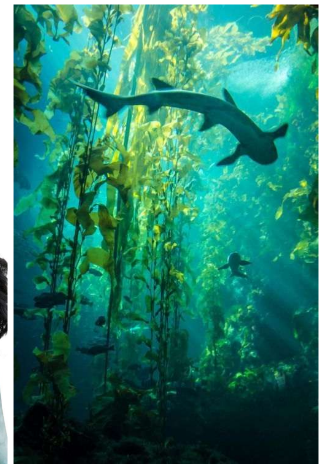
A desert planet: would have limited plant fibers, so maybe they have leather or woven grasses instead.
Extreme temperatures mean you need a layering system for both.
And water scarcity means you need materials that don't need washing.
So the environment creates the material palette directly.

What's available dictates what's possible

Ocean/Aquatic World:

- Abundant marine materials
  → Seaweed, shells, coral, marine animal hides
- Constant moisture
  → Waterproof or quick-drying essential
- Swimming primary movement
  → Minimal drag, doesn't impede motion





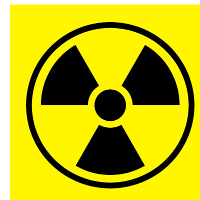
For an An ocean world:
There might be abundant marine materials, and the constant moisture means quick-drying or waterproof is essential,
and swimming as the primary movement means drag is a real problem you need to design with.

So biology and environment work together to point at a clear design direction

What's available dictates what's possible

High-Radiation Planet:

- Need radiation shielding
  → Metallic materials, dense weaves, reflective surfaces
- Harsh environment
  → Durable, doesn't degrade quickly under radiation
- All local life adapted
  → Might harvest materials with natural radiation resistance



On a High-radiation planet: might mean you need shielding so metallic, dense, reflective.
Everything needs to survive constant radiation exposure. And if local life is already adapted to it, will means whatever materials exist probably have radiation-resistance built in.
So here you see that the world hands you the logic.

Same biology, different cultures

Warrior Culture:

- Prioritize protection and mobility
- Materials:
  Durable, damage-resistant
- Design:
  Practical, minimal decoration (or decoration = battle honors)

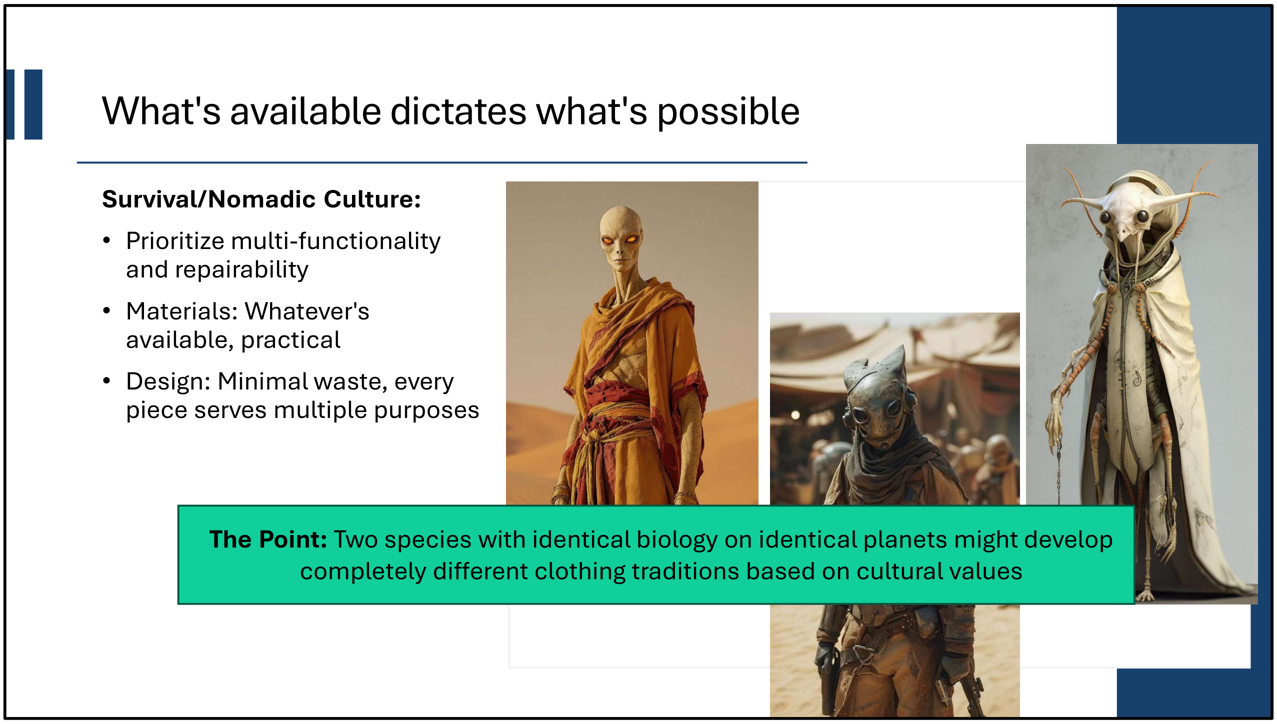
Even With Same Biology, same planet, but different cultures, creates completely different clothing.
A warrior culture prioritises protection and mobility.
Decoration reads as battle honours. Everything is functionality
But even this can be designed differently based on all the other design directions as you can see in the images.

What's available dictates what's possible

Spiritual/Religious Culture:

- Prioritize ritual significance and symbolism
- Materials:
  Culturally meaningful (sacred plants, blessed metals, symbolic colours)
- Design:
  Ceremonial elements, traditional patterns

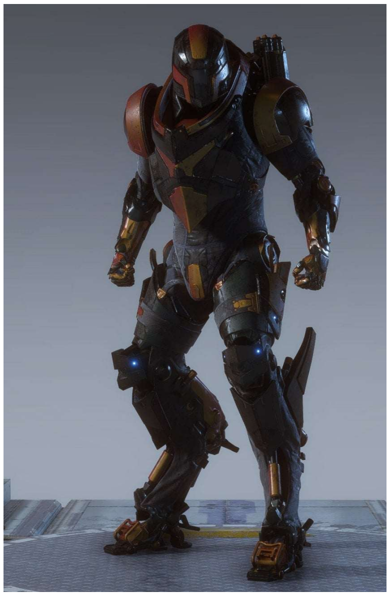
For A spiritual culture that prioritises ritual and symbolism.
You might want to design culturally meaningful materials, like ceremonial patterns.

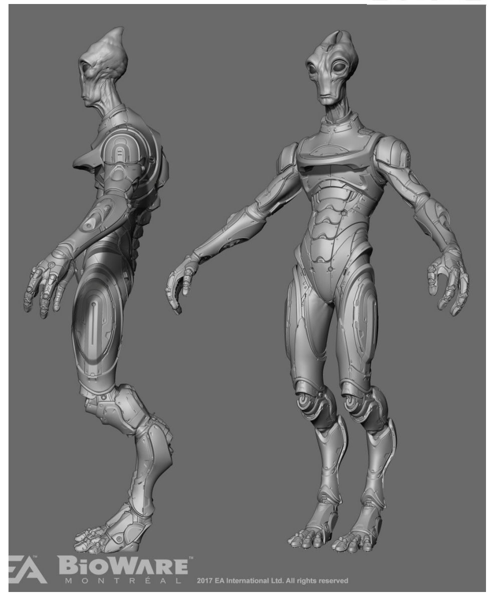


And then A nomadic survival culture prioritises multi-function and repairability. Nothing is purely decorative and every piece serves multiple purposes. Culture is a design variable just like biology yand environment.

--My point here is that two species with identical biology on the same planet might develop completely different clothing traditions based on cultural values. As you can also see on earth.

Game examples, good and bad



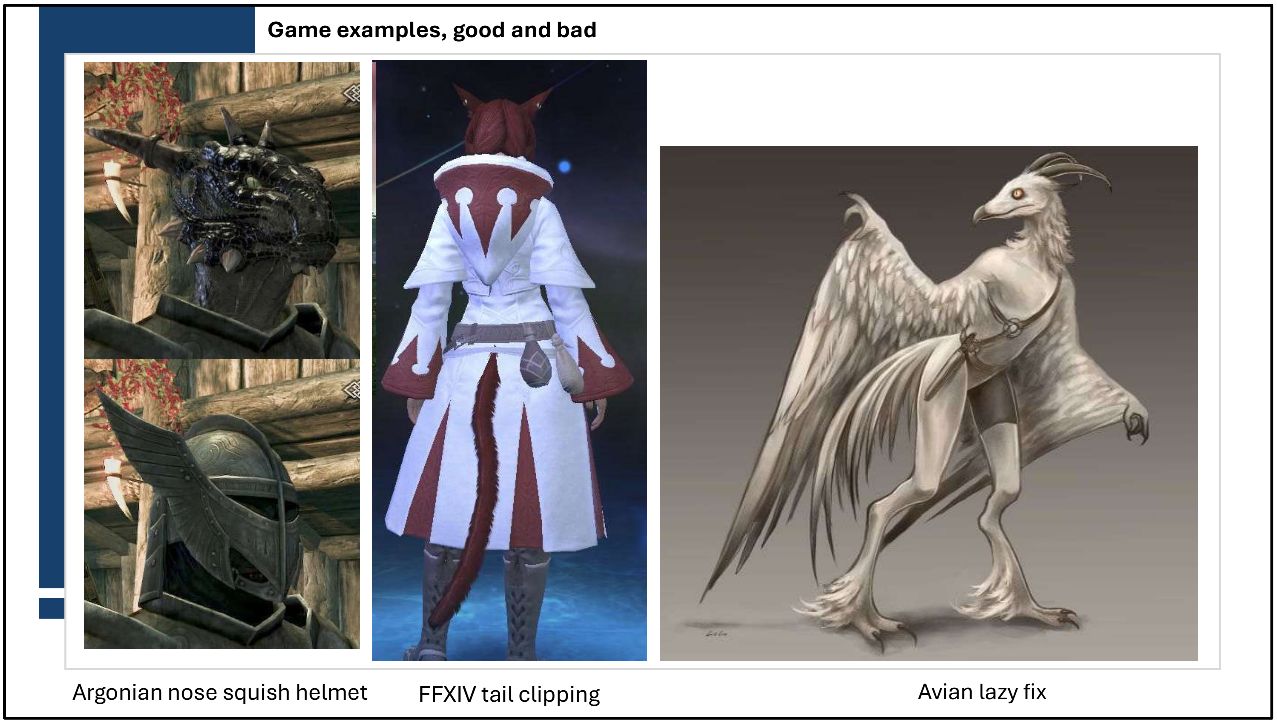


Turian armour

Krogan armour

Salarian armour

Here are some good Examples well done alien armour from mass effect
The outfits work around body differences, like karapace
Its designed with their biology in mind, the planet they come from, the natural protection they already have or miss.



And here some BAD Examples: Generic "Alien Armor“
The argonian from skyrim that gets his skull reconstructed when wearing helmets.
Tails that clip through jackets or capes
Or lazy fixes like avoiding the design problem completely

Quick question

Based on what we just discussed;

- What is one thing you'd now consider when designing alien costume that you wouldn't have thought of before?

Before the activity — what's one thing you'd now consider when designing alien costume that you wouldn't have before?

Lets start with the activity
We're going to analyse two different aliens together

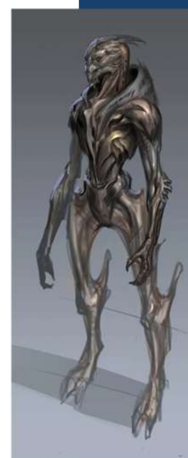
Activity: Alien Species Analysis

We're analyzing 2 species:

- **Metkayina** (*Avatar*) - Aquatic Na'vi
- **Turians** (*Mass Effect*) - High-radiation adapted

For each species, we'll discuss:

- What's their biology?
- What's their environment?
- What would they NEED clothing to do?
- What materials might they invent/use?
- What does canon show them wearing? Does it make sense?



Two species: Metkayina from Avatar: The Way of Water, and Turians from Mass Effect.

For eachone we’ll look at biology, environment, what they'd need clothing to do, and what materials they'd use, and whether what's in canon actually makes sense.

Species 2 - Metkayina species analysis

Metkayina: The Aquatic Na’vi

Biology:

- Aquatic adaptations (broader tails for propulsion, webbed hands)
- Larger forearms (swimming power)
- Turquoise/green skin with countershading (aquatic camouflage)
- Can hold breath for extended periods (free-diving)

Environment:

- Pandoran ocean atolls and coral reefs
- Warm tropical marine climate
- Live in woven marui pods above water
- Constant water exposure

So these are the Aquatic Na'vi. They have Broader paddle tails, webbed hands, larger forearms for swimming, countershading for aquatic camouflage, and can do extended free-diving.
Their environment is Ocean atolls, warm tropical climate, they live in woven pods above water. And theyre In and out of water constantly so not fully aquatic, not fully land-based.

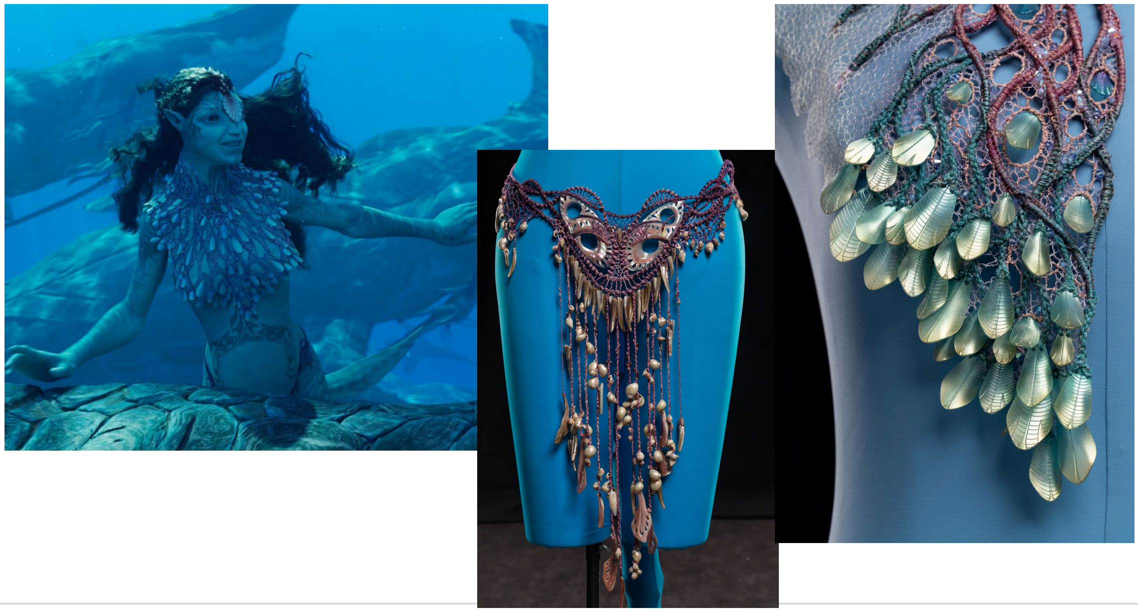
Species 2 - Metkayina species analysis

Discussion Questions:

1. They spend most of their time in water. What's the biggest clothing challenge?
2. They have broader hands and paddle tails for swimming. How does clothing avoid impeding this?
3. They only have access to marine materials (seaweed, shells, coral). What can they NOT make?
4. Polynesian inspiration is clear - but how does Pandoran ocean biology change it?

So lets discuss this

1. They spend most of their time in water. What's the biggest clothing challenge?
2. They have broader hands and paddle tails for swimming. How does clothing avoid impeding this?
3. They only have access to marine materials (seaweed, shells, coral). What can they NOT make?
4. Polynesian inspiration is clear - but how does Pandoran ocean biology change it?



What They Actually Invented:
Woven seaweed/marine plant fibers (lightweight, water-compatible, locally abundant)
Shells and coral (decoration, tools, cultural significance)
Minimal processed textiles (warm climate + swimming = little need)
Minimal coverage (reduces drag in water, prevents overheating on land)
Quick-drying natural fibers (grass skirts, woven elements)
Jewelry/decoration rather than garments (shells, coral, tattoos as primary "costume")
Open construction (doesn't restrict swimming motion or trap water)
Culturally inspired by Polynesian traditions (adapted to alien biology and marine materials)

Species 2 - Turian species analysis

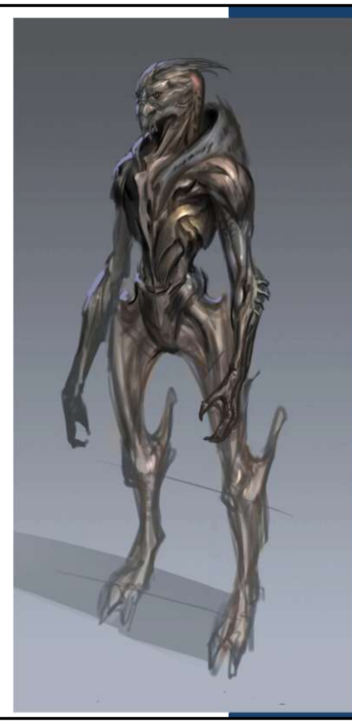
Turians: The Radiation-Adapted Avians

Biology:

- Metallic carapace with thulium (reflective, radiation-resistant)
- Carapace is NOT armor (doesn't stop weapons)
- Avian-predator features (head crests, taloned hands, jutting leg bones)
- Dextro-amino acid biology (incompatible with most galactic life)

Environment:

- Palaven: Weak magnetic field = high solar radiation
- Otherwise Earth-like (tropical, humid, temperate)
- "Silver world of fortresses" (metallic aesthetic)
- All native life has metallic adaptations



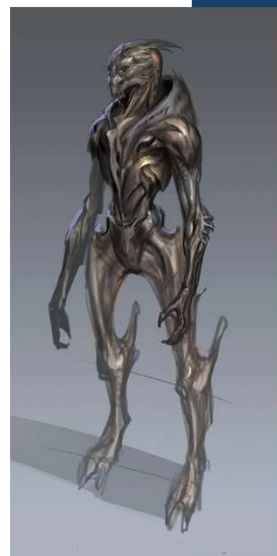
The turian biology is More complex. They have Metallic carapace with thulium a reflective and radiation-resistant, an adaptation to their homewords weak magnetic field.

But the carapace doesn't stop weapons. Its Environmental protection, not combat protection. So Head crests, taloned hands, jutting leg bones, and their biochemistry are incompatible with most galactic life.

Species 2 - Turian species analysis

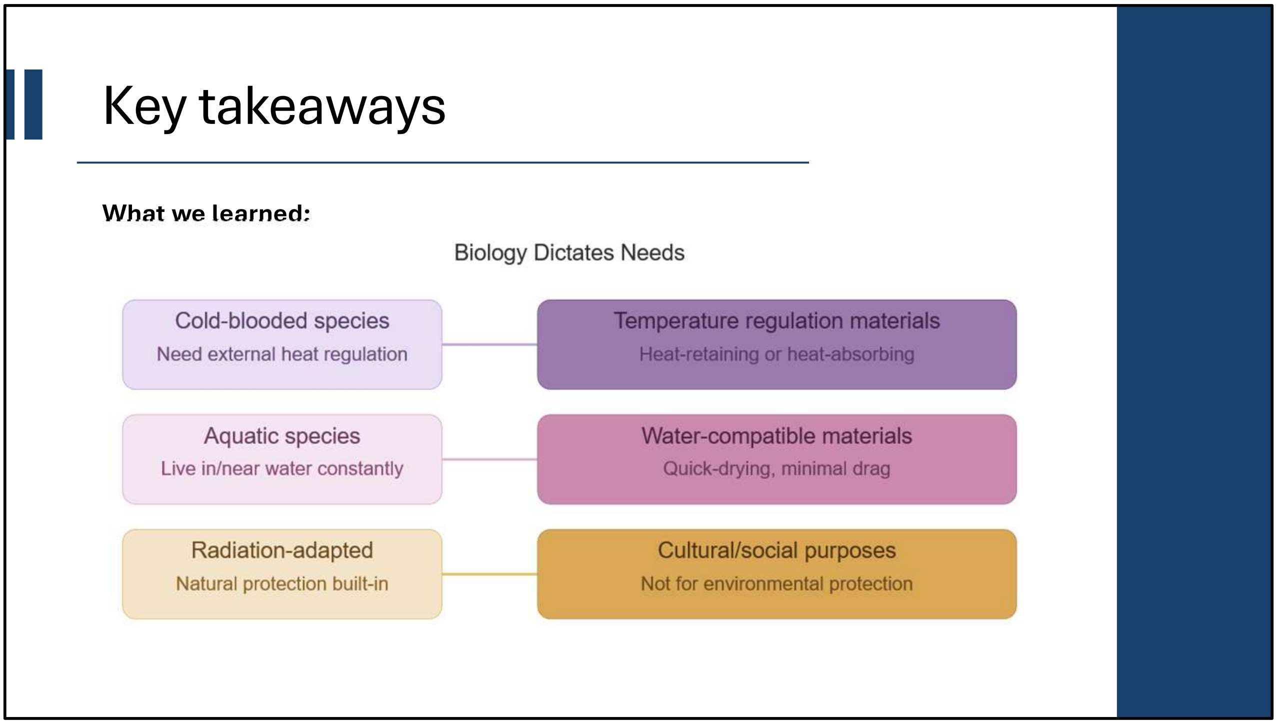
Discussion Questions:

1. They already have radiation protection from their carapace. So why wear clothing at all?
2. Their head crests, leg bones, and shoulder plates are very prominent. How does clothing work around these?
3. They're a militaristic culture that values service. How would that influence their textile/clothing development?
4. They can't eat most foods due to dextro-amino acids. Does biological incompatibility affect material choices?

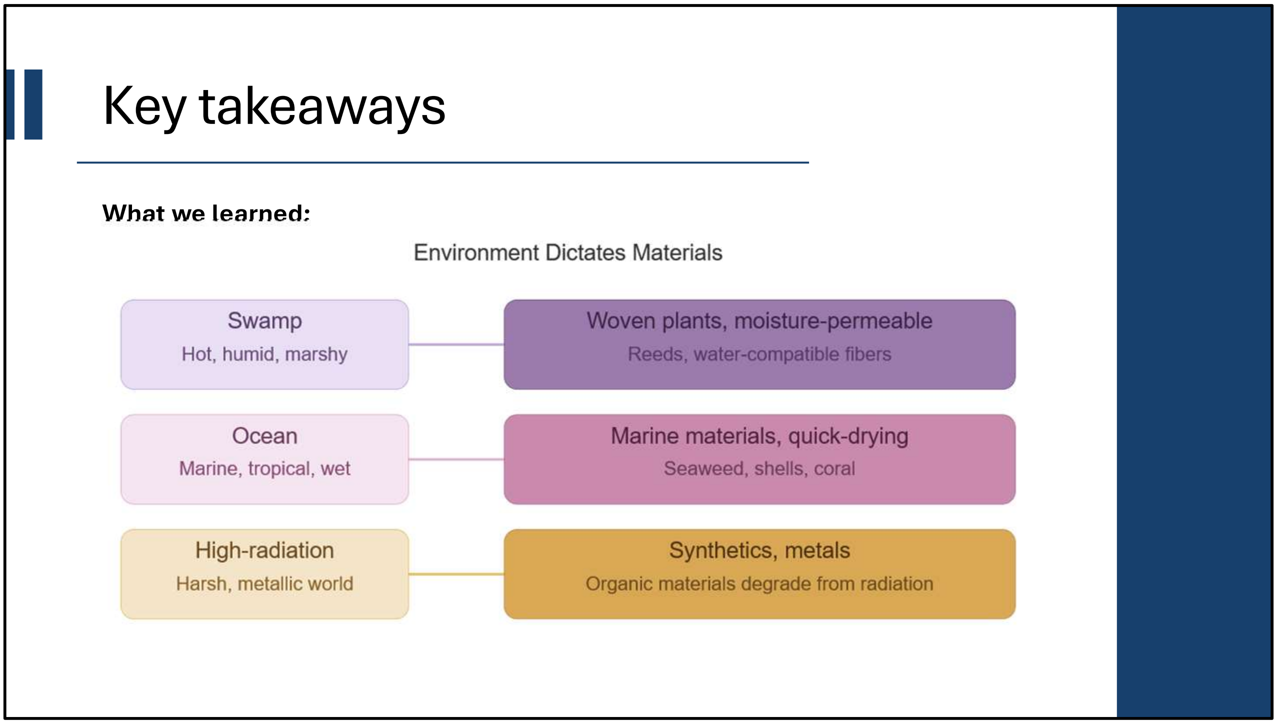


This one is trickier - clothing isn't for environmental protection, so what IS it for? Lets go through the questions together

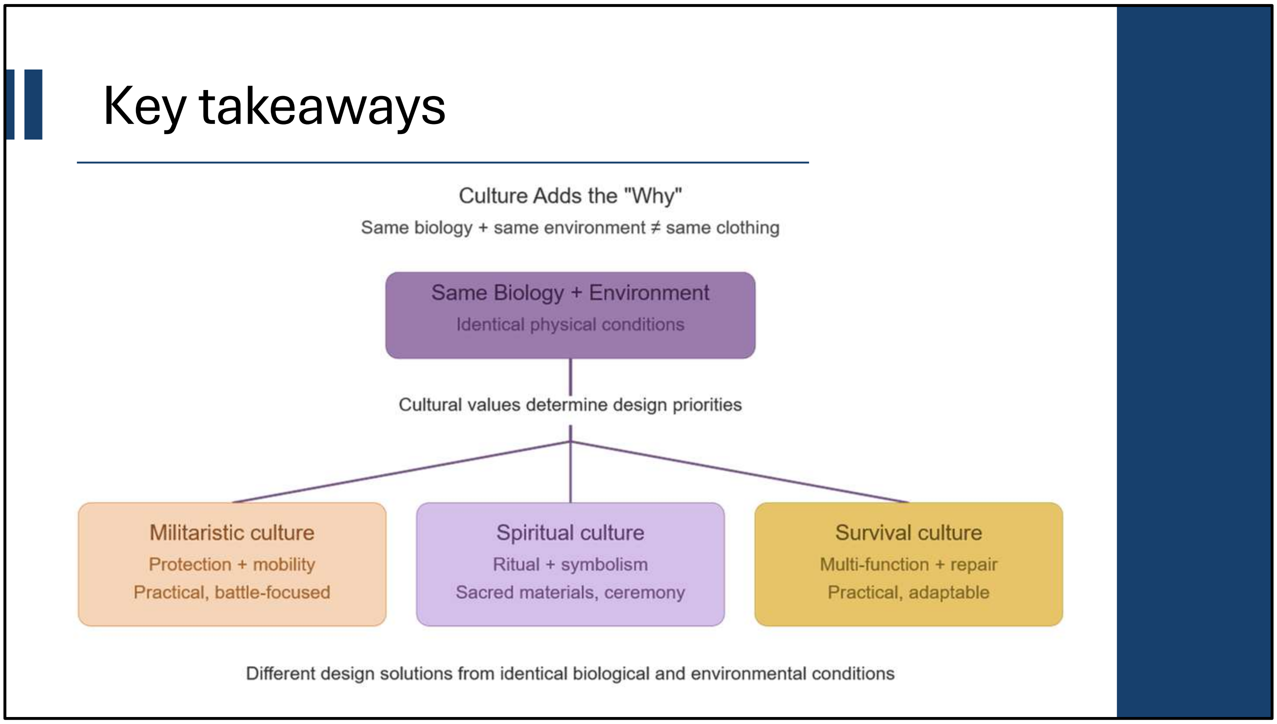
What They'd Likely Develop:
Purpose of Clothing (NOT environmental protection):
Cultural identification (colony markings, military rank, service record)
Military function (carrying equipment, tactical advantage, unit cohesion)
Social structure (displaying hierarchy, role, achievements)
Modesty/cultural norms (even if not biologically necessary)
Accommodation for carapace (shoulder plates, head crests, leg bones don't restrict)
Military aesthetic (structured, practical, service-oriented culture)
Metallic materials (complement natural carapace, culturally resonant)



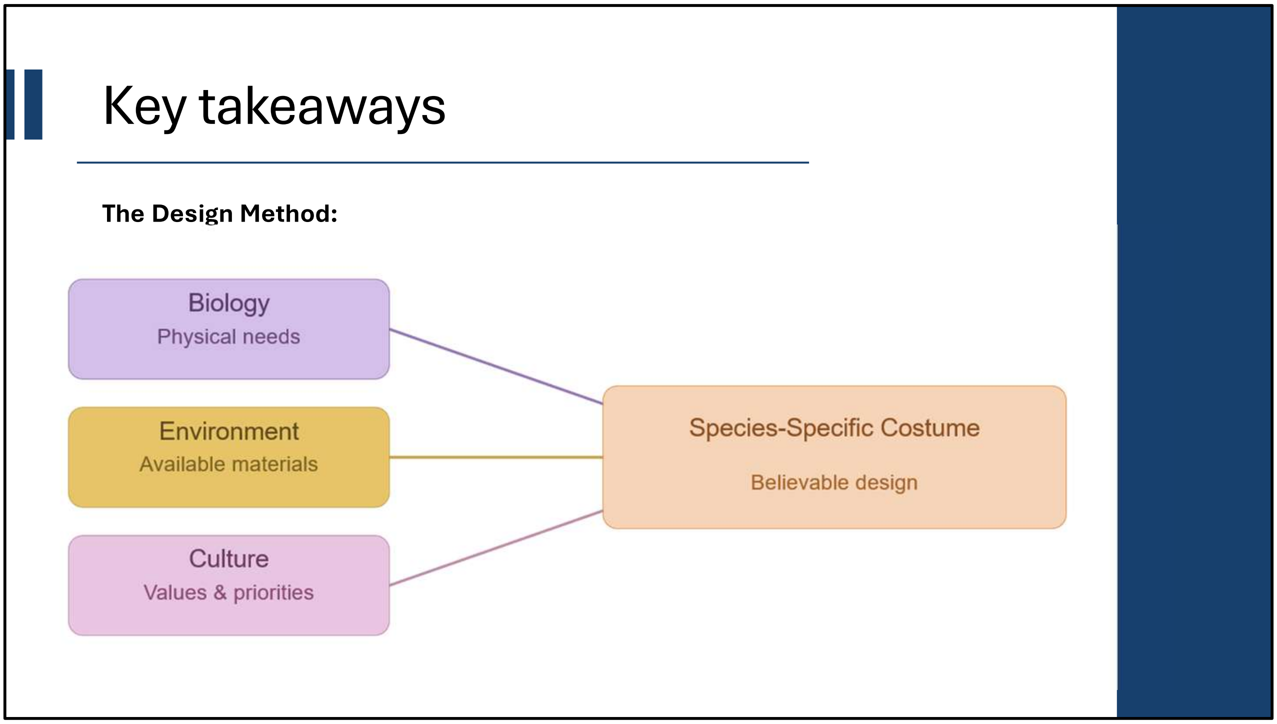
Alright so lets look at our key takeaways
Biology dictates the needs
A cold blooded species needs temperature regulation
An aquatic species needs water compatible materials
And radiation adapated species needs clothing for cultural or social purposes as environmental protection is less neccessary



Then the environment dictates what materials we have available
A swamp could contain woven plants and water compatible fibers
An ocean would give marine materials like seaweed, shells and coral
And a harsh high radiation environment would give synthetics and metals as organics degrade



And we learned that culture adds the “why” to our designs. Can be the same biology and environment but not the same clothing
The cultural values determine the design priorities as we’ve seen with the differences between militaristic culture, spiritual and survival cultures.
All different design solutions for identical conditions



And then to wrap this up the design method
You start with biology, what is the biological need
Then the environment, what materials are available
And finally the culture, how do values and priorities influence the design

And this all together creates a species specific costume grounded in believable design.

Check-in question

Based on analysing these two species:

- What would you now consider when designing costume for a non-human that you wouldn't have thought about before?

Based on analysing these two species:
what's one thing you'd consider when designing non-human costume that you genuinely wouldn't have thought of before today?

Time for the work review

Show your work!

- Anything you want to share?
- Get feedback on?

- Voluntary ☺

Prompts:
- What's working?
- What biology/environment considerations might improve it?
- How could you apply today's principles?

Anyone working on costuming right now? Can you share something you want to talk through and get feedback on?
Can you tell us whats working, what biology/environment might improve it? And how would you apply todays principles?

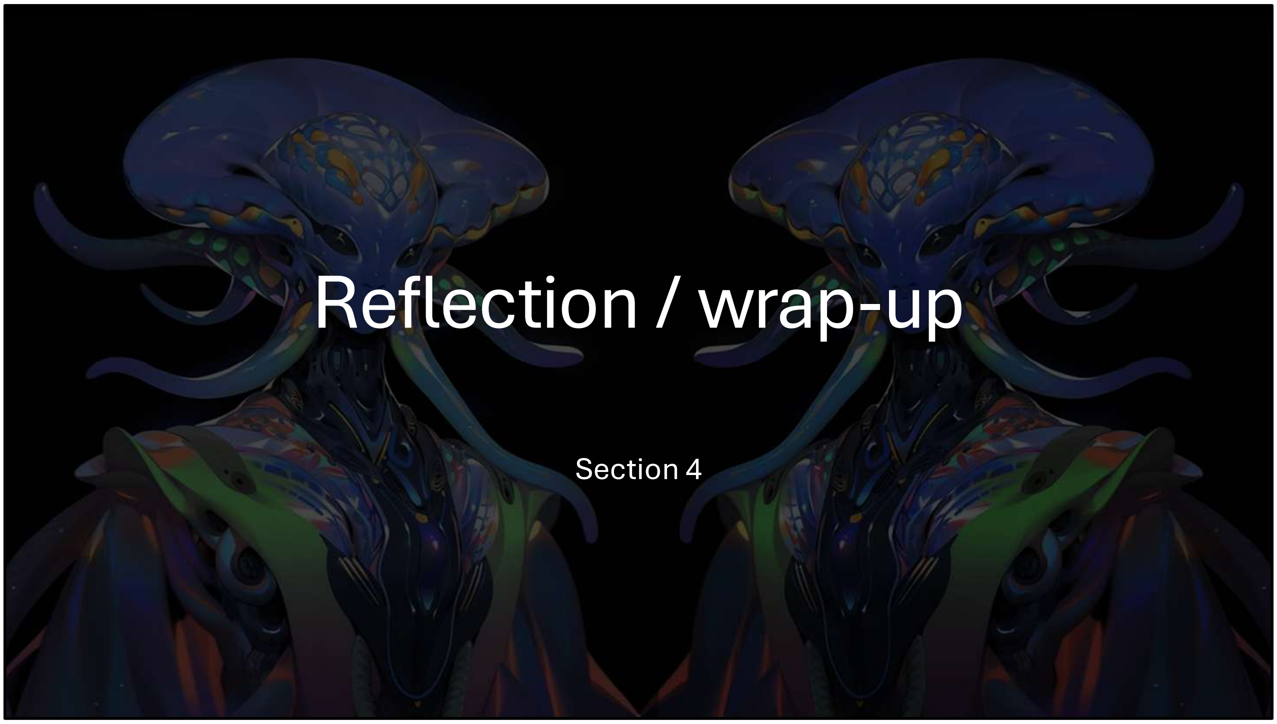
Show your work!

Tell us:

- What species/creature is this?
- What's their biology/environment?
- What were you trying to communicate with their costume?
- What's challenging you?

We'll discuss:

- Biology/environment alignment
- Material choices (do they make sense for this species?)
- Cultural coding (what does the costume communicate?)
- Practical suggestions



Then time for the reflection and wrap up

Key takeaways from today

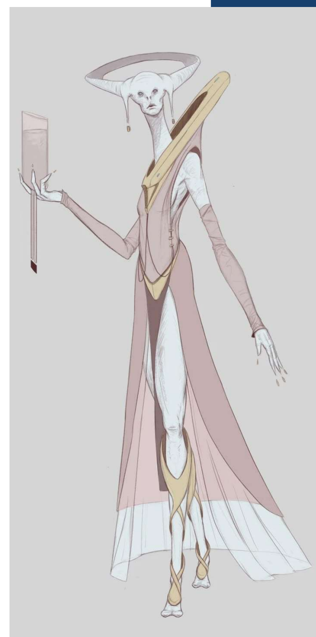
✅ **Biology determines material NEEDS**
(thermoregulation, protection, movement)

✅ **Environment determines material AVAILABILITY**
(can only use what exists in their world)

✅ **Culture determines design CHOICES**
(what they VALUE influences what they make)

✅ **"Human clothing with tweaks" is lazy**
- think deeper about species-specific logic

✅ **There's no single right answer**
- justify your choices with biological/environmental/cultural reasoning



Here are some key takeaways you can look at once more

Reflection prompt

Think about a non-human character (yours or from a game).

Ask yourself:

- Does their costume make sense for their biology?
- What would THEY invent based on their environment?
- What cultural values would shape their clothing traditions?

If theres time